Applied ML – COMS4995
HW1 – From Dirty Data to Predictive Models – Titanic Survival
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Problem statement
Predict whether a passenger survived or not.

Dataset Description

Variable	Definition	Key
survival	Survival	0 = No, 1 = Yes
pclass	Ticket class	1 = 1st, 2 = 2nd, 3 = 3rd
sex	Sex	
Age	Age in years	
sibsp	# of siblings / spouses aboard the Titanic	
parch	# of parents / children aboard the Titanic	
ticket	Ticket number	
fare	Passenger fare	
cabin	Cabin number	
embarked	Port of Embarkation	C = Cherbourg, Q = Queenstown, S = Southampton

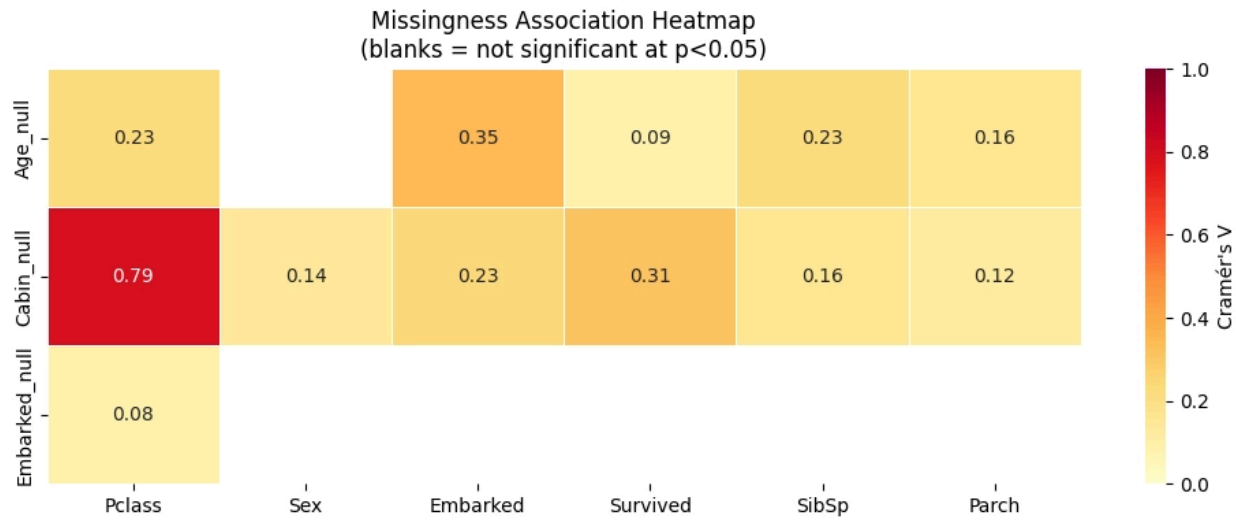
Data Cleaning

The following columns have missing values:
1. Age
2. Cabin
3. Embarked

Initial analysis compared survival rates between rows where values were present vs. missing:
- Age – Moderate difference, but nothing glaring. Revisit later.
- Cabin – Presence of a cabin value resulted in a delta of +36.7% for survival.
- Embarked – While a large delta exists, note that n=2 for the null cases. Rows were dropped.

Analyzing Correlation Between Missing Data and Other Variables — Cramér's V

Cramér's V is a method to calculate the correlation between two categorical variables. A heatmap was generated for each missing value column.
Key finding: Cabin_null × PClass has a higher Cramér's V, warranting further investigation.



Cabin – Flag

Over 90% null rate for PClass ∈ {2, 3}.
This suggests structural information in the data. Cabin is a luxury, so it makes sense that the majority of PClass = 1 passengers have a cabin value. Passengers in classes 2 and 3 largely do not.

Rather than imputing or dropping, a binary flag was created to capture whether a Cabin value was present. This preserves the signal identified above.

Pclass	null_count	null_rate	total
1	40	0.185185	216
2	168	0.913043	184
3	479	0.975560	491

Embarked – Drop

There are only 2 data points where Embarked is missing. Since this is well under 1% of the data, these rows were dropped to resolve the missing values.

has_cabin	count	survival_rate
No Cabin	687	0.299854
Has Cabin	204	0.666667

has_cabin value counts:

has_cabin	
0	687
1	204

Age – Impute

Initial approach was to impute Age, as it is a continuous variable rather than discrete (like Embarked and Cabin). The key question was how to impute it.

Age × Embarked

Port Q (Queenstown) has the highest age null rate, and also the highest PClass = 3 representation.

However, mean age by Embarked port does not differ meaningfully (roughly within 2 years). This does not provide a useful imputation basis.

```
          count  survival_rate  mean_fare  median_fare  mean_age  pct_female  \
Embarked
C           168          0.554     59.954        29.70    30.815       0.435
Q            77          0.390     13.276         7.75    28.089       0.468
S           644          0.337     27.080        13.00    29.445       0.315

          age_null_rate
Embarked
C                 0.226
Q                 0.636
S                 0.140

— Pclass distribution per port —
Pclass        1      2      3
Embarked
C         0.506  0.101  0.393
Q         0.026  0.039  0.935
S         0.197  0.255  0.548
```

Age × SibSp & Parch

Intuitively, age clusters should appear depending on whether a passenger had family on board. For example, young children (ages 0–10) are expected to have parents and therefore a non-zero Parch value.

Correlation analysis confirms that SibSp, Parch, and PClass all show meaningful correlations with Age.

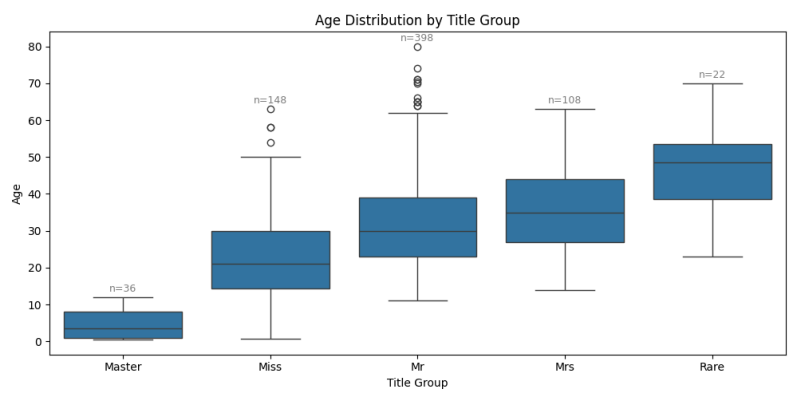
Approach 1 — Impute Age Using Linear Regression (SibSp, Parch, PClass)

These features show significant correlations with Age. The logic: children tend to travel with parents (non-zero Parch), and wealthier passengers (PClass = 1) are likely older on average. We use Linear Regresion model to impute the age.

```
SibSp: -0.307
Parch: -0.188
Fare: 0.093
Pclass: -0.366
```

Approach 2 — Impute Age by Title Group and Pclass

Titles extracted from passenger names show clear age signal. For example, 'Master' was a title given to young boys, producing a distinct age cluster. We use a linear regression model, using OHE on the Title Groups.



Approach 3 — Impute Age by Combining Approach 1 and 2, using Linear Regression

A combined approach using both regression features and title groups was also evaluated.

Approach Chosen – Approach 3

Final imputation method selected based on highest R^2 adjusted value.

```
Comparison (R² / Adj R²):
  Numeric only (SibSp, Parch, Pclass)      R² = 0.220   Adj R² = 0.217   (p=3)
  Title + Pclass                           R² = 0.386   Adj R² = 0.382   (p=5)
  Combined (this model)                    R² = 0.400   Adj R² = 0.394   (p=7)
```

Data Cleaning Resulting Columns:

```
Columns: ['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age',
'SibSp', 'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked', 'Title',
'Age_null', 'has_cabin', 'title_group']
```

Feature Engineering

Available transformations: normalization, standardization, log-scaling, one-hot encoding (OHE), and feature creation.

```
— family_size —
family_size
1     535
2     161
3     102
4      29
5      15
6      22
7      12
8       6
11      7
Name: count, dtype: int64

is_alone: 60.2% of passengers

— Correlation with Survived —
family_size: 0.018
is_alone: -0.206
```

Analyzing the frequency of our new feature, as well as calculating correlation with each new feature with survival rate

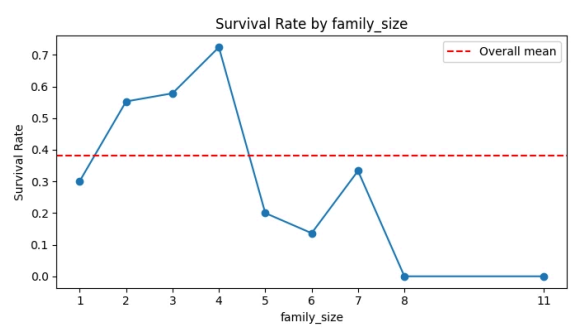
Planned Transformations

4. PassengerId — Drop (correlation with Survived ≈ −0.005)
5. Title — One-Hot Encoding
6. Sex — One-Hot Encoding
7. Age — Normalize
8. SibSp + Parch → FamilySize; create IsAlone when FamilySize = 1
9. Fare — Log transform (expected right skew)
10. Embarked — One-Hot Encoding

Family Size & IsAlone

FamilySize = SibSp + Parch
IsAlone = (FamilySize == 1)
IsAlone was created because the marginal effect of going from alone to having any family member is likely different from incremental increases in family size (2→3, 3→4, etc.).

IsAlone showed a correlation of −0.206 with survival — significant. FamilySize correlation was weaker.



Diving Deeper into Family Size

Survival rates show a clear drop-off for FamilySize > 4. A numerical value does not correctly capture this dynamic.

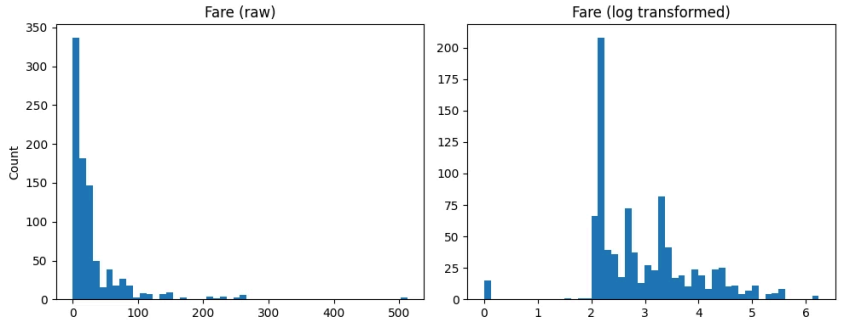
Final decision: bin FamilySize into [alone, medium, large] and apply One-Hot Encoding. Resulting features: fam_size_medium, fam_size_large.

Features to Encode

- Embarked — One-Hot Encoding
- Sex — One-Hot Encoding

Features to Transform

- Fare — Log transform (right-skewed distribution confirmed visually)
- Age — Normalize



Features to Drop

- PassengerId — No correlation with survival
- Name — Titles extracted; no signal in raw names
- Cabin — has_cabin flag already extracted
- Ticket — No meaningful signal

Resulting Data after Feature Engineering

```
['Survived', 'Pclass', 'Sex', 'Age', 'Fare', 'Age_null', 'has_cabin',
'fam_size_medium', 'fam_size_large', 'Embarked_Q', 'Embarked_S',
'Title_Miss', 'Title_Mr', 'Title_Mrs', 'Title_Rare']
```

Model Comparison

The following models were trained and evaluated:

11. Mixed Naive Bayes
12. Linear Regression
13. Lasso
14. Ridge

Listing a truncated list of our model accuracies, selecting only best for each Ridge and Lasso

Model	Train Acc	Test Acc
Mixed NB	0.7834	0.7528
Linear Regression	0.8340	0.8258
Lasso (α=0.001)	0.8383	0.8202
Ridge (α=1.0)	0.8340	0.8258

Mixed Naive Bayes

This model combines Bernoulli NB and Gaussian NB. Bernoulli NB handles categorical features; Gaussian NB handles numerical features. This hybrid approach was chosen because the dataset contains both types.

Results with Varying Alpha / Laplace Smoothing

No meaningful difference in accuracy was observed between alpha values. This is likely because Laplace smoothing is most useful for sparse data — which this dataset is not.

```
alpha=1.0   Training accuracy: 0.7829
alpha=0.01  Training accuracy: 0.7829
```

Linear Regression with Threshold

Since all features were transformed to numerical values, implementation was straightforward. Linear Regression outputs a value in [0, 1]. A threshold of 0.5 was applied to produce binary classifications.

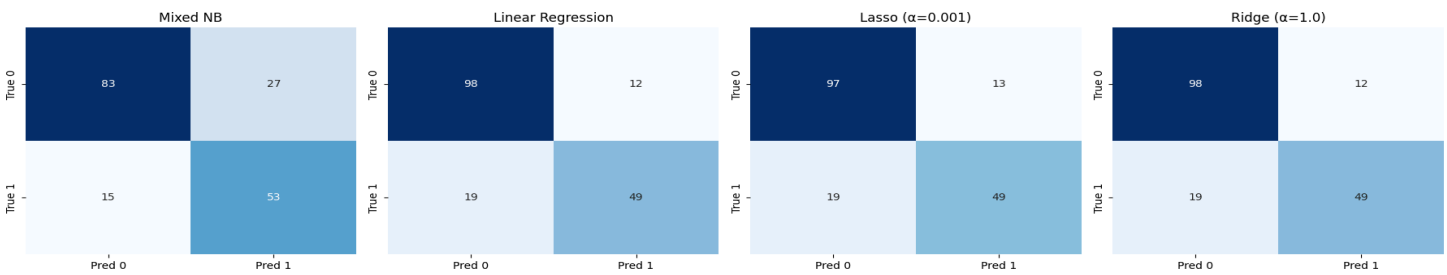
Confusion Matrices

Accuracy Comparison

Data was split 80/20 using stratified sampling (random_state=42)

Corresponding Metrics from Confusion Matrix

	TP	FP	FN	TN	Recall	Specificity	Precision	NPV	Accuracy	F1
Mixed NB	53.0	27.0	15.0	83.0	0.7794	0.7545	0.6625	0.8469	0.7640	0.7162
Linear Regression	49.0	12.0	19.0	98.0	0.7206	0.8909	0.8033	0.8376	0.8258	0.7597
Lasso (α=0.001)	49.0	13.0	19.0	97.0	0.7206	0.8818	0.7903	0.8362	0.8202	0.7538
Ridge (α=1.0)	49.0	12.0	19.0	98.0	0.7206	0.8909	0.8033	0.8376	0.8258	0.7597



Discussion

Model Results Summary

Mixed Naive Bayes

- Highest recall — catches the most actual survivors, but over-predicts survival
- Highest NPV — when the model predicts death, it is most likely correct
- Poor specificity and precision (~60–70%)
- Leans heavily toward predicting survival relative to other models
- F1 score is lowest of all models, reflecting this imbalance

Linear Regression, Lasso, and Ridge

Result variance is slim amongst these three models.

<u>Why didn't regularization improve performance?</u>
Lasso: Objective is to zero out irrelevant features. It did not help because all engineered features were relevant.

Title_Mr	14.76
Title_Miss	14.24
Title_Rare	13.90
Sex	13.73
fam_size_large	13.45
Title_Mrs	12.96
Age	6.33
has_cabin	2.74
Pclass	2.32
Embarked_Q	1.61
fam_size_medium	1.31
Fare	1.26
Embarked_S	0.76
Age_null	0.64

Ridge: Most helpful when model weights are disproportionately large. Weight analysis showed a fairly distributed spread — no disproportionate outliers — explaining Ridge's minimal impact.

Image to the right are the factors and cumulative weights (Sum = 1) our Linear Regression Model used:

F1 Score: Linear Regression / Lasso / Ridge scored higher than Naive Bayes by approximately 0.04.

Why Naive Bayes Underperformed

Naive Bayes assumes conditional independence among features. This assumption is violated throughout the dataset. For example, Title vs. Age, FamilySize vs. Age, and PClass vs. Age all show meaningful relationships. These violations explain the relatively lower performance. Linear Regression does not make this assumption.

Limitations

A test accuracy of ~83% is fairly good, but further gains may be possible – and also bounded by inherent data constraints:

1. Inherent signal loss: Age imputation and other missing data handling introduces noise.
2. No systematic rescue process: Decisions made in the moment involved human judgment, which introduces inconsistency in whichever heuristic they may have adopted.
3. Supply constraints: Even with consistent prioritization, limited lifeboats meant not all prioritized passengers could be saved.

AI Tool Attribution

Tools used: Claude and Gemini.

Both were used for generating all code, visualizations, and debugging. One-off questions included: how to find correlation for categorical variables (Cramér's V) and how to build a Naive Bayes model that handles both categorical and numerical data (Mixed NB).

I contributed everything not related to code execution: directing the analysis, deriving insights from results, and writing the full report.